MARIAM HAKRO

DATA ANALYST | COMPUTER SCIENCE STUDENT

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EDUCATION

Sukkur IBA University
Mirpurkhas Campus
BS Computer Science
2022 – Present | CGPA 3.06/4.0

Ibn-e-Rushed Girls College
HSC (Pre-Engineering)
2017 – 2019 | Grade: A

Govt. Girls High School
Fruit Farm, Mirpurkhas
SSC (Science Group)
2015 – 2017 | Grade: A

TECHNICAL SKILLS

- Python (Django)
- HTML, CSS, JavaScript
- Pandas, NumPy, EDA
- MySQL, PostgreSQL, SQLite
- Frontend & Backend Development
- Networking Fundamentals
- Cybersecurity Awareness
- MS Office & Documentation

SOFT SKILLS

- Leadership
- Teamwork
- Project Management
- Communication
- Critical Thinking
- Time Management

LANGUAGES

- English (Fluent)
- Urdu (Fluent)
- Sindhi (Fluent)
- Siraiki (Basic)

PROFILE

Motivated Computer Science student with hands-on experience in data analysis, cybersecurity fundamentals, and full-stack web development. Skilled in applying machine learning and data visualization to support data-driven decisions, with strong leadership and analytical ability and a growing interest in applying technology to secure, efficient banking operations.

WORK EXPERIENCE

Project Lead & AI Data Analyst
Data Analysis Remote Internship — Excelerate (Remote)

- Analyzed historical learner application data using exploratory data analysis (EDA) and time-series analysis to identify seasonal trends and peak periods.
- Developed machine learning models to forecast application demand and identify high-performing opportunity categories.
- Built interactive dashboards in Looker Studio to communicate insights and support data-driven business decisions.

Data Visualization Associate
Data Handling & Analysis Internship — Excelerate (Remote)

- Worked with structured datasets using Pandas and NumPy for data organization and analysis.
- Cleaned and processed raw data to generate actionable insights for reporting.

PROJECTS

QIDS — Hybrid Intrusion Detection System
Final Year Project — Classical & Quantum Machine Learning for Cyber Intrusion Detection

- Developed a Hybrid Intrusion Detection System (QIDS) using classical and quantum machine learning models to detect and classify cyberattacks.
- Built a real-time network monitoring module using Scapy with a Django-based dashboard for live threat visualization.
- Implemented AI-based attack detection, risk scoring, security alerting, and automated PDF report generation.
- Benchmarked 5 classical and 2 quantum ML models on the CICIDS2017 dataset, comparing performance across precision, recall, F1-score, and execution time.

EventToEase — Digital Event Management System
System Developer (Group Project)

- Contributed to development of a digital event booking platform, handling user data and system workflow.
- Ensured structured information management across the application, applying concepts transferable to banking-style digital service platforms.

Secure Client-Server Communication System
Networking Project — Academic (GitHub)

- Developed a client-server communication model simulating secure data transfer between systems.
- Implemented a structured request-response mechanism demonstrating IP communication, data transmission, and connectivity handling.